

# AI/ML intern DAY 15

## Garment Warping & Fitting Research for Virtual Try-On Systems

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### 1. Introduction

Virtual Try-On (VTON) systems aim to digitally fit garments onto a user's body while maintaining realistic appearance, proper alignment, and natural garment behaviour. One of the most important stages in a Virtual Try-On pipeline is garment warping and fitting.

Garment warping refers to the process of transforming a clothing image or 3D garment model so that it matches the body shape, posture, and proportions of a target user. Fitting techniques ensure that garments appear naturally worn rather than simply overlaid on top of the body.

Modern AI-based Virtual Try-On systems use computer vision, pose estimation, body segmentation, geometric transformations, and deep learning models to achieve realistic fitting results.

This research explores major garment warping and fitting techniques including:

- Thin Plate Spline (TPS)
  - Flow-Based Warping
  - CP-VTON
  - VITON-HD
  - Geometric Matching Module (GMM)
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### 2. Objectives

The primary objectives of this research are:

- Understand garment warping techniques
  - Study clothing adaptation to body shapes
  - Analyse fitting methods used in Virtual Try-On systems
  - Compare major garment alignment architectures
  - Identify strengths and limitations of each approach
  - Recommend suitable solutions for future implementation
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### 3. Importance of Garment Warping

In Virtual Try-On systems, garments cannot simply be pasted onto a person's image. Different users have:

- Different body shapes
- Different body sizes
- Different poses
- Different camera angles

Without garment warping, clothing appears unrealistic and misaligned.

Garment warping helps:

- Align clothing with body posture
  - Adjust garment dimensions
  - Preserve garment texture
  - Improve realism
  - Reduce visual artifacts
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### 4. Thin Plate Spline (TPS)

#### Overview

Thin Plate Spline (TPS) is a geometric transformation technique widely used for image warping. TPS deforms garments smoothly based on body landmarks and control points.

The algorithm calculates transformations that minimize distortion while adapting clothing to the target body.

#### Workflow

Garment Image

↓

Control Point Extraction

↓

Body Landmark Matching

↓

TPS Transformation

↓

Warped Garment Output

#### Advantages

- Smooth transformations
- Computationally efficient
- Easy implementation
- Good alignment quality

## Limitations

- Limited handling of complex deformations
- Performance decreases with extreme poses

## Applications

- Virtual Try-On systems
- Medical image registration
- Object alignment
- Computer vision tasks



**Figure 1:** TPS Garment Warping Architecture

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## 5. Flow-Based Warping

### Overview

Flow-Based Warping uses optical flow estimation to determine pixel-level movement between garment and body images.

Instead of relying only on geometric transformations, flow-based approaches estimate how every pixel should move to match the target body.

### Workflow

Garment Image  
↓  
Flow Estimation Network  
↓  
Pixel Displacement Prediction  
↓

Garment Warping

↓

Fitted Output

### **Advantages**

- Handles complex deformations
- Better texture preservation
- Accurate fitting

### **Limitations**

- Higher computational cost
- Requires larger training datasets

### **Applications**

- Advanced Virtual Try-On systems
- Video-based fitting
- Motion adaptation

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## **6. CP-VTON**

### **Overview**

CP-VTON (Characteristic-Preserving Virtual Try-On Network) is one of the earliest deep-learning-based Virtual Try-On frameworks.

The system preserves garment characteristics while adapting clothing to the user.

### **Architecture**

Person Representation

+

Clothing Image

↓

Geometric Matching Module (GMM)

↓

TPS Warping

↓

Try-On Network

↓

Final Output

### **Features**

- Characteristic preservation
- TPS-based garment warping

- End-to-end architecture
- Improved garment alignment

### Advantages

- Lightweight architecture
- Good visual results
- Preserves garment textures

### Limitations

- Limited resolution
- Struggles with complex poses

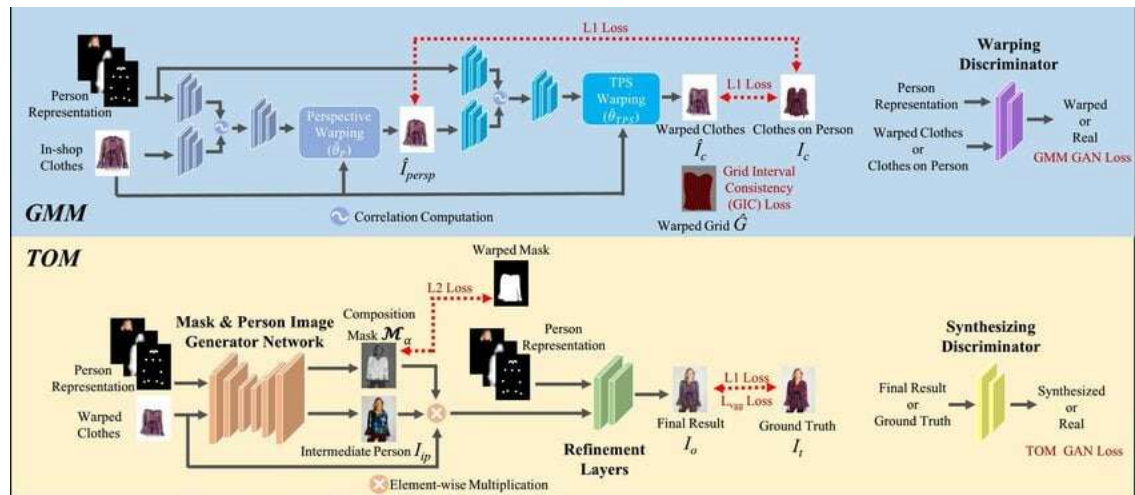


Figure 2: CP-VTON Architecture

## 7. VITON-HD

### Overview

VITON-HD is an improved high-resolution Virtual Try-On framework designed to generate realistic garment fitting results.

The system improves texture preservation and garment details compared to earlier VTON approaches.

### Architecture

User Image

↓

Pose Estimation

↓

Body Segmentation

↓

Garment Warping

↓

Image Generation Network



High-Resolution Output

### Features

- High-resolution output
- Better texture quality
- Improved realism
- Enhanced garment details

### Advantages

- State-of-the-art results
- Better visual quality
- Realistic garment fitting

### Limitations

- High computational requirements
- Longer inference time

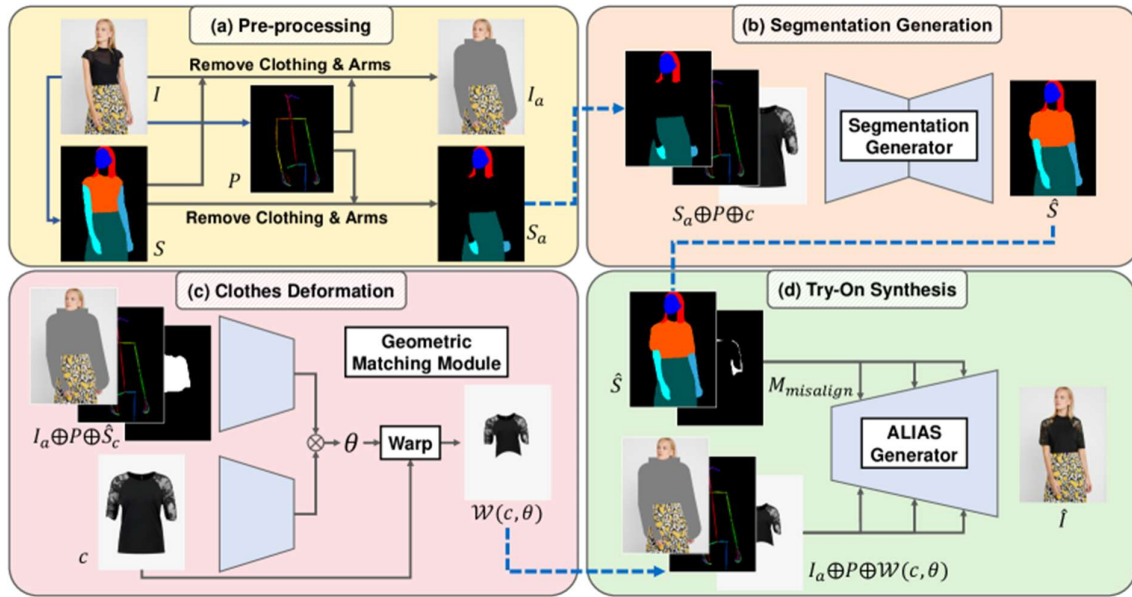


Figure 3: VITON-HD Architecture

## 8. Geometric Matching Module (GMM)

### Overview

Geometric Matching Module (GMM) is a neural network component responsible for aligning garments with target body shapes.

GMM predicts transformation parameters that map garment images to user body regions.

**Workflow**

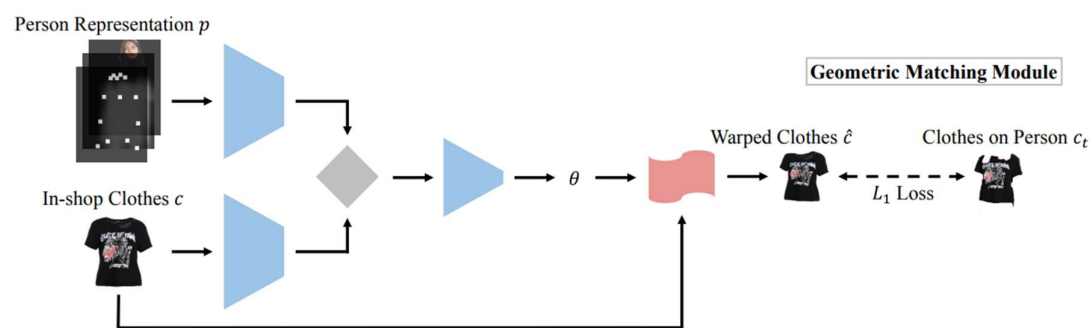
Garment Features  
+  
Body Features  
↓  
Feature Matching  
↓  
Transformation Prediction  
↓  
Garment Alignment

**Advantages**

- Accurate alignment
- Learns body-garment relationships
- Flexible fitting

**Limitations**

- Requires training data
- Sensitive to segmentation quality



**Figure 4:** Geometric Matching Module (GMM)

**9. Technology Comparison**

| Technology         | Realism   | Complexity | Speed  | Texture Preservation |
|--------------------|-----------|------------|--------|----------------------|
| TPS                | Medium    | Low        | High   | Medium               |
| Flow-Based Warping | High      | High       | Medium | High                 |
| CP-VTON            | High      | Medium     | Medium | High                 |
| VITON-HD           | Very High | High       | Medium | Very High            |
| GMM                | High      | Medium     | High   | High                 |

## 10. Garment Warping Workflow

The complete garment fitting pipeline follows:

User Image



Body Detection



Pose Detection



Body Segmentation



Garment Feature Extraction



Geometric Matching



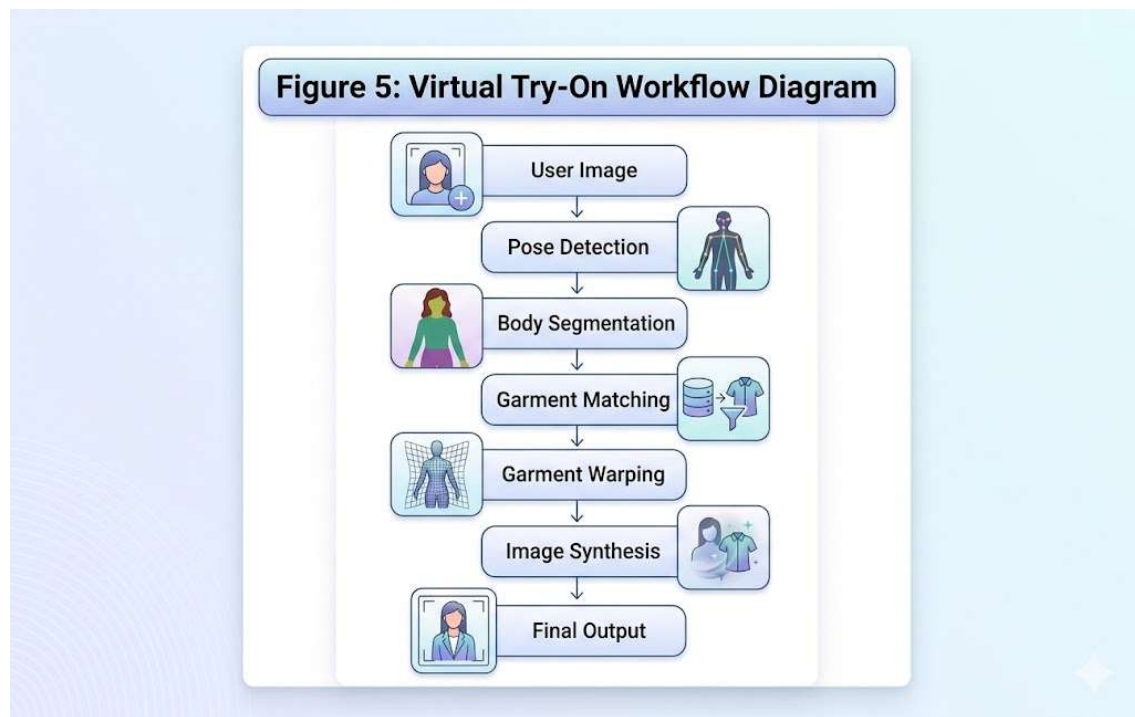
Garment Warping



Image Synthesis



Virtual Try-On Output



**Figure 5:** Virtual Try-On Workflow Diagram



## **11. Research Findings**

### **Best Geometric Warping Method**

TPS

Reason:

- Fast
- Lightweight
- Easy implementation

### **Best Deep Learning Approach**

CP-VTON

Reason:

- Characteristic preservation
- Good alignment quality

### **Best High-Resolution Output**

VITON-HD

Reason:

- Realistic fitting
- High visual quality

### **Best Overall Alignment Module**

GMM

Reason:

- Learns garment-body relationships
- Flexible adaptation

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## **12. Challenges Identified**

Several challenges were identified during the study:

### **Body Shape Variations**

Different body types require adaptive fitting methods.

### **Pose Variations**

Extreme poses create alignment difficulties.

### **Texture Preservation**

Maintaining garment details remains challenging.

### **Occlusion Handling**

Arms, hair, and accessories may block clothing regions.

### **Real-Time Processing**

High-quality warping often increases computational requirements.

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### **13. Recommended Solution**

For future Virtual Try-On development:

GMM + TPS + VITON-HD

Benefits:

- Accurate garment alignment
  - High-resolution output
  - Better texture preservation
  - Improved realism
  - Scalable architecture
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### **14. Future Improvements**

Potential improvements include:

- 3D garment fitting
  - Real-time cloth simulation
  - Physics-based garment deformation
  - Multi-view fitting
  - Personalized body measurements
  - AI-based fabric behaviour modelling
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### **15. Conclusion**

This research explored major garment warping and fitting techniques used in modern Virtual Try-On systems. TPS, Flow-Based Warping, CP-VTON, VITON-HD, and GMM were analysed based on realism, fitting quality, computational complexity, and scalability.

The findings indicate that TPS provides efficient geometric alignment, GMM improves body-garment matching, and VITON-HD delivers the highest visual quality. A combination of GMM, TPS, and VITON-HD is recommended for future Virtual Try-On systems due to its balance between realism, performance, and scalability.